

Cloud & Controller Managed Access Point. (NAV-319-VA)



Dual-band 2x2:2 MU-MIMO WiFi-6 Cloud Managed Access Point

The NAV-319-VA is a WiFi-6, two-radio, cloud-managed access point designed for next-generation deployments. It offers high performance, enterprise-grade security, and centralized management, delivering a maximum 3Gbps aggregate frame rate.



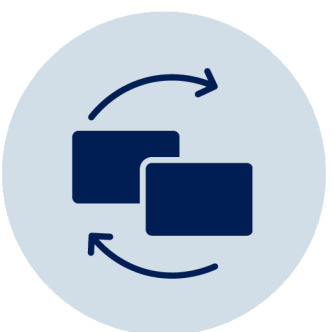
High Data Rate

Supports a total aggregated link speed of 3000Mbps. Offers 600Mbps in the 2.4GHz band and 2400Mbps in the 5GHz band for a great browsing experience.



MU-MIMO Technology

Features Multi-user Multiple Input and Multiple Output (MU-MIMO) technology, allowing efficient spectrum use and supporting a high number of concurrent users.



Cloud Managed & Plug-n-Play

Fully self-configurable. Once powered on, it downloads configurations from the cloud immediately. Includes support for guest access and VLAN isolation.

Aggregated Speed

3000Mbps

High Throughput Performance

Client Capacity

100+

Concurrent Users Supported

The NAV-319-VA delivers the high throughput, reliability, and flexibility required by the most demanding business applications like voice and high-definition streaming video. Designed for general-purpose, next-generation deployments, its cloud architecture makes provisioning, configuration, and monitoring simple and error-free, ensuring the device becomes operational within minutes of powering on.

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Product Overview

The **NAV-319-VA** is a modern Wi-Fi 6 (802.11ax) access point designed for businesses. It features two radios (dual-band) and cloud management capabilities, making it ideal for indoor locations that need high performance and easy central management.

This device delivers a maximum combined speed of 3Gbps (3000Mbps) across its 2.4GHz and 5GHz bands. It is designed to handle demanding applications, such as voice calls and high-definition video streaming, ensuring reliability even in challenging environments.

Using advanced OpenWiFi technology, the NAV-319-VA is managed through Indio's cloud platform. This cloud-based system makes setting up, configuring, and monitoring remote devices simple and error-free.

Key Points

- Can be used as a ceiling mount, wall mount/desktop device.
- Aggregate data rate of 3Gbps.
- Self-configuring, self-healing mesh.
- Inbuilt 5dBi Omnidirectional Antennas.
- IEEE 802.11a/b/g/n/ac/ax compliant.
- 2.4GHz and 5GHz concurrent radios.
- Remote management via HTTP, HTTPS, Secure SSH, Telnet, and SSL.
- Configuration file backup and restore by TFTP or FTP, HTML.
- Rich system information –AP status, station status, event logs.
- IPv4 and IPv6 dual stack support.

Key Features

High-Speed Performance

- Delivers 3000Mbps aggregated speed (600Mbps on 2.4GHz + 2400Mbps on 5GHz).
- Powered by a Qualcomm 1GHz Quad-Core CPU for stable, high-throughput processing.

Smart Connectivity

- Features 2x2:2 MU-MIMO technology for efficient simultaneous data transmission.
- Supports 100+ concurrent active devices without performance drops.

Cloud & Mesh Ready

- Zero-touch, plug-and-play configuration via the Indio Cloud Controller.
- Supports self-healing mesh networking for extended coverage and redundancy

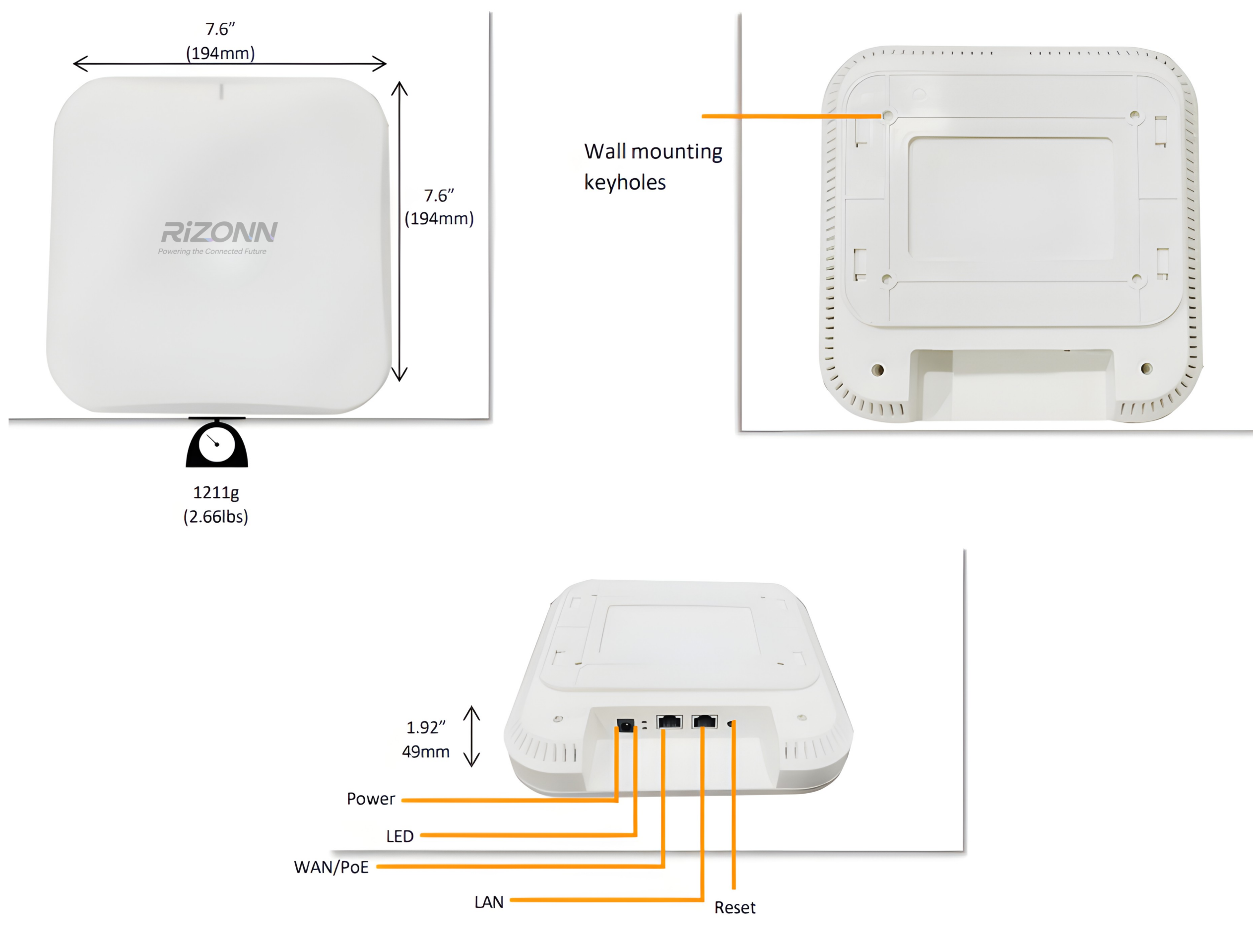
Enterprise Security

- Comprehensive security with WPA3, VLAN isolation, and Guest Access modes.
- Includes Captive Portal, SMS/OTP authentication, and deep packet inspection features.

Easy Management & Monitoring

- Supports comprehensive remote management via HTTP, HTTPS, Secure SSH, Telnet, and SSL.
- Integrated troubleshooting utilities, including syslog, ping, traceroute, and speed-test.
- Provides rich system information with detailed logs for AP status, station status, and network events.

A robust dual-band WiFi-6 access point designed for indoor environments. It combines 3Gbps speeds with seamless cloud management, making it ideal for high-density deployments requiring reliable voice and HD video streaming.



Access Point Deployment

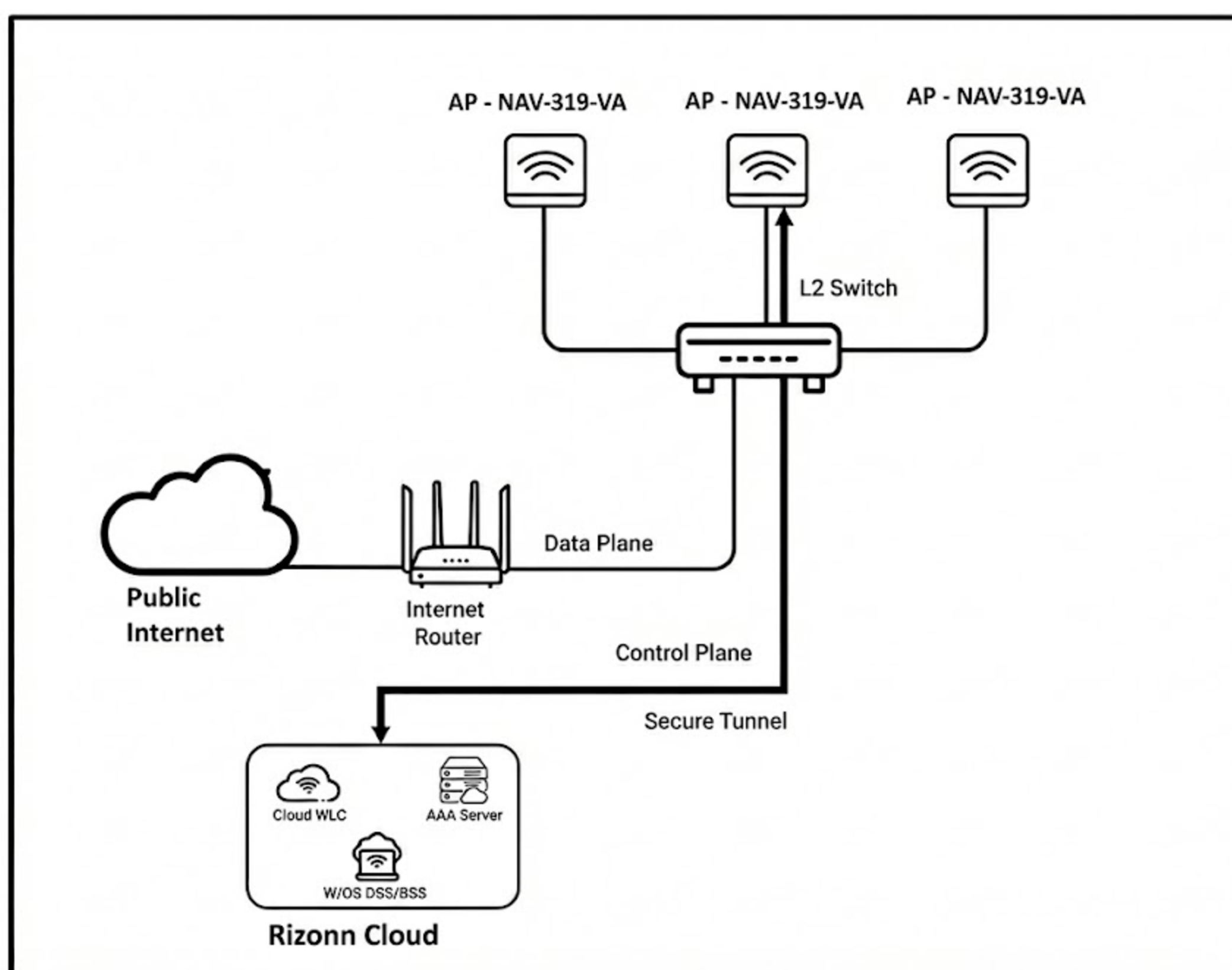


Figure 1. RiZonn AP installation with RiZonn Cloud Controller.

Technical Specifications

Category	Feature	Specification
Hardware	Throughput & Users	3000Mbps; 100+ users
	Radios	2.4GHz: 802.11b/g/n/ax; 5GHz: 802.11a/n/ac/ax
	Chipset	Qualcomm 1GHz Quad-Core CPU with dedicated NSS
	Memory / Flash	512MB RAM / 32MB Flash
	Ports (Interfaces)	1x WAN (10/100/1000/2500 Mbps); 1x LAN (10/100/1000 Mbps)
	Antenna	Inbuilt 5dBi omnidirectional antennas
	Power Consumption	24W max
	Power Source	PoE (802.3af/at) or Proprietary 48V
	Output Power	2.4GHz: 24dBm
	Reset Button	1
Wireless	WiFi Standards	IEEE 802.11 a/b/g/n/ac/ax
	Supported Rates	2.4GHz: Max 600Mbps
	Peak Data Rates	2.4GHz: 573.5Mbps
	MIMO / Streams	2x2 MIMO (2 Spatial Streams)
	Frequency Bands	2.4GHz: 2400–2482 MHz
		5GHz: 5150–5250 MHz, 5250–5350 MHz, 5470–5725 MHz, 5725–5850 MHz
	Security	WPA/WPA2/WPA3
	SSIDs	8 Per Radio
	Client Capacity	128+ Concurrent devices
	Other Features	WMM, Power Save, Fast Roaming (802.11r/k/v), Hotspot Service, Captive Portal, WISPr
Networking	Operation Modes	WiFi Access Point, Mesh
	Management Platform	Cloud Controller Appliance managed & Stand-alone mode
	IP Stack	IPv4/IPv6
	VLAN Support	802.1Q (static and dynamic using Radius), Port based
	Authentication	RADIUS authentication, 802.1x (Auth & Supplicant), PSK, SAE
	Tunnels	VPN, GRE
	Network Policies	ACL, Rate Limiting, Device Fingerprinting
Software Features	Installation	Plug-n-Play (Zero Touch)
	Rate Limiting	Per Client, Per SSID, Dynamic Rate Limiting, MAC Authentication, MAC Filtering, Firewall
	Guest Portals	Cloud portals: Click-through, Data Capture, SMS/OTP, Payment Gateway, Social-Media, Voucher
	Troubleshooting	syslog, ping, traceroute, speed-test
	Time Sync	NTP
	Data Offload	Yes
	SNMP / Loc API	No

Environmental	Operating Temp	-20°C ~ 60°C
	Humidity	5% ~ 95%
	Mounting	Ceiling / Wall mount
Certifications	Safety	EN 60950-1
	Wireless	WPC
	Compliance	OpenWiFi Compliant, Passpoint Compliant

2.4GHz Table

Operating Band	Operating Mode	Data Rate	TX Power	RX Sensitivity
2.4 GHz	802.11b	1 Mb/s	27 dBm	-96 dBm
		2 Mb/s	27 dBm	-93 dBm
		5.5 Mb/s	27 dBm	-91 dBm
		11 Mb/s	27 dBm	-89 dBm
		6 Mb/s	27 dBm	-91 dBm
		9 Mb/s	27 dBm	-90 dBm
2.4 GHz	802.11g	12 Mb/s	27 dBm	-88 dBm
		18 Mb/s	26 dBm	-87 dBm
		24 Mb/s	26 dBm	-84 dBm
		36 Mb/s	26 dBm	-81 dBm
		48 Mb/s	25 dBm	-76 dBm
		54 Mb/s	25 dBm	-75 dBm
2.4 GHz	802.11n (HT20)	MCS0/8	27/27 dBm	-91/91 dBm
		MCS1/9	27/27 dBm	-88/88 dBm
		MCS2/10	26/26 dBm	-85/85 dBm
		MCS3/11	26/26 dBm	-82/82 dBm
		MCS4/12	25/25 dBm	-79/79 dBm
		MCS5/13	25/25 dBm	-75/75 dBm
		MCS6/14	25/25 dBm	-73/73 dBm
		MCS7/15	24/24 dBm	-70/70 dBm

2.4 GHz	802.11n (HT40)	MCS0/8	27/27 dBm	-89/89 dBm
		MCS1/9	27/27 dBm	-86/86 dBm
		MCS2/10	26/26 dBm	-83/83 dBm
		MCS3/11	26/26 dBm	-80/80 dBm
		MCS4/12	25/25 dBm	-77/77 dBm
		MCS5/13	25/25 dBm	-73/73 dBm
		MCS6/14	25/25 dBm	-72/72 dBm
		MCS7/15	24/24 dBm	-70/70 dBm

5GHz Table

5GHz Receive Sensitivity & Transmit Power

Operating Band	Operating Mode	Data Rate / MCS	TX Power (dBm)	RX Sensitivity (dBm)
5 GHz	802.11a	6 Mb/s	26 dBm	-90 dBm
		9 Mb/s	26 dBm	-87 dBm
		12 Mb/s	26 dBm	-86 dBm
		18 Mb/s	26 dBm	-85 dBm
		24 Mb/s	25 dBm	-84 dBm
		36 Mb/s	25 dBm	-79 dBm
		48 Mb/s	25 dBm	-74 dBm
		54 Mb/s	26 dBm	-71 dBm
5 GHz	802.11n (HT20)	MCS0/8	26/26 dBm	-88/-88 dBm
		MCS1/9	26/26 dBm	-85/-85 dBm
		MCS2/10	25/25 dBm	-83/-83 dBm
		MCS3/11	25/25 dBm	-79/-79 dBm
		MCS4/12	24/24 dBm	-76/-76 dBm
		MCS5/13	24/24 dBm	-72/-72 dBm
		MCS6/14	23/23 dBm	-71/-71 dBm
		MCS7/15	23/23 dBm	-69/-69 dBm
5 GHz	802.11n (VHT20)	MCS0/0	26/26 dBm	-88/-88 dBm
		MCS1/1	26/26 dBm	-86/-86 dBm
		MCS2/2	25/25 dBm	-83/-83 dBm
		MCS3/3	25/25 dBm	-79/-79 dBm
		MCS4/4	24/24 dBm	-77/-77 dBm
		MCS5/5	24/24 dBm	-75/-75 dBm
		MCS6/6	23/23 dBm	-72/-72 dBm
		MCS7/7	23/23 dBm	-70/-70 dBm
		MCS8/8	22/22 dBm	-67/-67 dBm

5 GHz	802.11n (HT40)	MCS0/8	26/26 dBm	-85/-85 dBm
		MCS1/9	26/26 dBm	-84/-84 dBm
		MCS2/10	25/25 dBm	-84/-84 dBm
		MCS3/11	25/25 dBm	-79/-79 dBm
		MCS4/12	24/24 dBm	-77/-77 dBm
		MCS5/13	24/24 dBm	-72/-72 dBm
		MCS6/14	23/23 dBm	-70/-70 dBm
		MCS7/15	23/23 dBm	-68/-68 dBm
5 GHz	802.11n (VHT40)	MCS0/0	26/26 dBm	-85/-85 dBm
		MCS1/1	26/26 dBm	-82/-82 dBm
		MCS2/2	25/25 dBm	-79/-79 dBm
		MCS3/3	25/25 dBm	-77/-77 dBm
		MCS4/4	24/24 dBm	-74/-74 dBm
		MCS5/5	24/24 dBm	-70/-70 dBm
		MCS6/6	23/23 dBm	-68/-68 dBm
		MCS7/7	23/23 dBm	-67/-67 dBm
		MCS8/8	22/22 dBm	-64/-64 dBm
		MCS9/9	22/22 dBm	-63/-63 dBm
5 GHz	802.11ac (VHT80)	MCS0/0	26/26 dBm	-83/-83 dBm
		MCS1/1	26/26 dBm	-81/-81 dBm
		MCS2/2	25/25 dBm	-79/-79 dBm
		MCS3/3	25/25 dBm	-76/-76 dBm
		MCS4/4	24/24 dBm	-73/-73 dBm
		MCS5/5	24/24 dBm	-70/-70 dBm
		MCS6/6	23/23 dBm	-67/-67 dBm
		MCS7/7	23/23 dBm	-66/-66 dBm
		MCS8/8	22/22 dBm	-62/-62 dBm
		MCS9/9	22/22 dBm	[cite_start]-60/-60 dBm

